

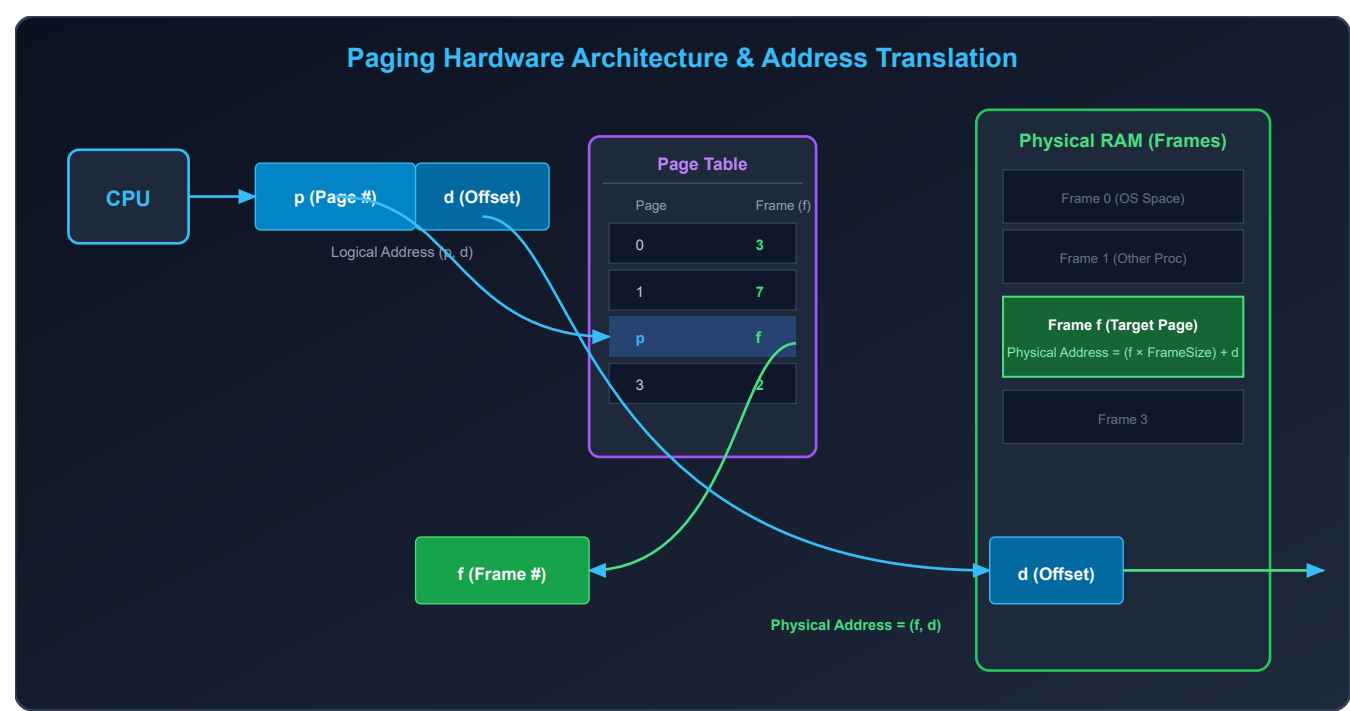
Module 04: Paging Architecture and Address Translation

1. Fundamental Paging Rules

Physical memory is split into fixed-size **Frames**; Logical space is split into identical-size **Pages**. Frame Size = Page Size = 2^n bytes. External fragmentation is completely eliminated!

2. Address Translation Formula

CPU generates (p, d). MMU accesses Page Table at index p to retrieve Frame Number f. Physical Address = (f × FrameSize) + d.



3. Key Formulas & Summary Table

Parameter	Mathematical Formula	Example (32-bit CPU, 4KB page)
Offset Bits (d)	$\log_2(\text{Page Size})$	$\log_2(4096) = 12$ bits
Page Bits (p)	$\text{Logical Bits (m)} - d$	$32 - 12 = 20$ bits
Total Pages	2^p	$2^{20} = 1\text{M}$ pages
Frame Bits (f)	$\log_2(\text{RAM Size} / \text{Frame Size})$	$\log_2(512\text{MB} / 4\text{KB}) = 17$ bits
Internal Frag	Worst: $\text{PageSize} - 1$; Avg: $\text{PageSize} / 2$	Worst: 4095 bytes; Avg: 2048 bytes

4. Solved Address Translation Numerical

Given: Page Size = 4096 bytes, Page 5 maps to Frame 12.

Logical Address: 20500.

1. Page Number $p = \text{floor}(20500 / 4096) = 5$.
2. Offset $d = 20500 \bmod 4096 = 20$.
3. Physical Address = $(12 \times 4096) + 20 = 49152 + 20 = \mathbf{49172}$.